BUSINESS ANALYTICS CASE STUDY

Targeted Marketing Strategy for Bank Term Deposit Subscriptions

Using conversion-driver analysis and a cost-savings model to prioritize who the bank calls, and how often.

Timothy Kehinde Promise

AnalystLab Africa Data Analytics Internship • Week 5

Business Problem

A bank runs outbound telemarketing campaigns to sell term deposits. Every contact attempt costs real call-center time and money, yet only a fraction of customers contacted actually subscribe.

Calling every customer with equal effort wastes budget on low-probability contacts and reduces overall campaign efficiency.

THE QUESTION THIS ANALYSIS ANSWERS

Which customer segments and campaign conditions produce the highest conversion probability, and how much could the bank save by prioritizing those segments instead of contacting everyone equally?

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REFRAMED AS

A Targeting & Cost-Efficiency Problem

Not just “who subscribed”
but “who should we
call first, and how often”

Dataset Overview

Bank Marketing Dataset (UCI / Kaggle)

11,162	17	0	47.4%
Customer contact records	Original fields: demographic, financial, campaign	Missing values or duplicate rows	Dataset conversion rate (artificially balanced)

TWO CORRECTIONS MADE BEFORE ANALYSIS

1. Target variable is artificially balanced at 47.4%. Real-world bank telemarketing converts at roughly 11%. All cost figures in this study are reweighted against the realistic 11% baseline.

2. Call duration is only known after a call ends, so it cannot inform who to call beforehand. Excluded from all targeting logic; used only as a call-quality observation.

Analytical Approach

Four stages, each verified in PostgreSQL before moving to the next

1

Audit

Row count, nulls, duplicates, target balance, and “unknown” placeholder values checked first.

2

Clean

Built prior_contact_flag from three overlapping unknown fields; standardized job and month labels.

3

Query

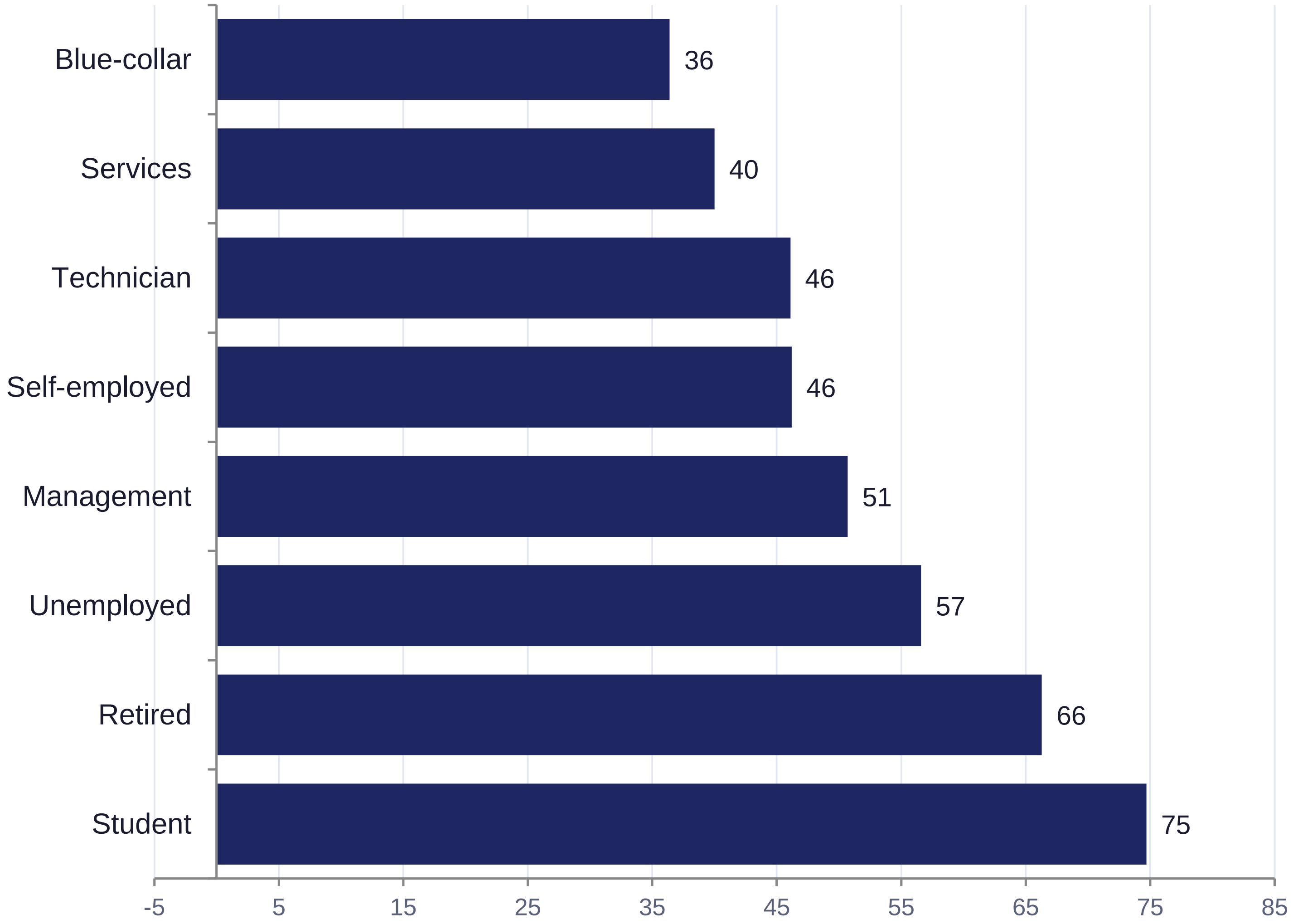
12 SQL queries answering specific business questions across every customer attribute.

4

Model

Two targeting approaches compared, then translated into a dollar-based cost-savings model.

Key Finding: Conversion Varies Sharply by Occupation



THE PATTERN

Students and retirees convert nearly twice the rate of blue-collar workers, despite blue-collar being the campaign's second-largest segment by contact volume.

Likely driver: segments with fewer competing financial obligations (loans, dependents) have more disposable income to lock into a term deposit.

This same pattern repeats across marital status, loan status, and age.

Targeted Calling Is 42% More Cost-Efficient

Cost per conversion, reweighted to a realistic 11% baseline conversion rate

TARGETED SEGMENTS

$30.78

per conversion

2,612 contacts

16.2% realistic conversion rate

~424 estimated conversions

NOT TARGETED

$53.20

per conversion

8,550 contacts

9.4% realistic conversion rate

~804 estimated conversions

Why the Obvious Move Is the Wrong One

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It is tempting to say: stop calling the Not Targeted group entirely. But that group still produces 804 of the total 1,228 estimated conversions, nearly two-thirds of total volume. Dropping it means giving up most of the pipeline, not improving it.

THE ACTUAL RECOMMENDATION: CAP CONTACT ATTEMPTS

53.4% → 26.8%

Conversion rate falls from
1 contact to 7+ contacts

$32,775

Estimated savings from
capping contact at 3 calls
(6,555 excess calls avoided)

562 customers

converted only after their 3rd call (16% of this segment's conversions). Pilot the cap before full rollout to confirm real impact.

Business Insight

The segments this campaign currently calls the most — married customers, blue-collar workers, and customers with no prior contact history — are also the segments that convert the least.

A large share of the campaign's call volume and cost is concentrated in its least productive segments, rather than being weighted toward the customers most likely to say yes.

ON THE MULTI-VARIABLE TEST

A more detailed targeting list (job + prior contact + loan status + contact count) reached a higher top conversion rate, but on groups as small as 30–70 people, too small to act on with confidence. The simpler, statistically reliable two-variable list was chosen deliberately over the marginally higher but unstable one.

Recommendations

1

Prioritize targeted segments first

Call qualifying job + prior-contact segments early in the cycle — they convert at 42% lower cost per conversion.

2

Cap contact attempts at 3 calls

Applies to lower-converting segments only. Estimated $32,775 saved; pilot before full rollout.

3

Reassess offers for loan holders

Housing/personal loan holders convert at roughly half the rate — test a lower minimum deposit or flexible terms.

4

Audit the “unknown” contact channel

Converts at less than half the rate of cellular/telephone — likely a logging gap, not a behavioral signal.

5

Validate against real campaign data

This dataset is artificially balanced; confirm segment-level lift against the bank's own records before budgeting.

Conclusion

A small set of characteristics — occupation, prior contact history, and existing loan obligations — consistently explain who is more likely to subscribe. The campaign's current calling pattern does not reflect this: its highest-volume segments are also its least efficient.

Shifting call priority toward high-converting segments, while capping wasted repeat contact elsewhere, gives the bank a realistic, measurable path to a more cost-efficient campaign — without abandoning any part of its customer base.

Thank you

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